

BUSINESS ANALYSIS AND VALUATION

Valuation of Private Companies

KEY DIFFERENCES: PUBLIC VS PRIVATE VALUATION

Feature	Public Companies	Private Companies
Financial Data	Readily available, standardized quarterly and annual reports, SEBI filings	Limited availability, often not standardized, may require adjustments for owner expenses
Market Price	Observable daily stock prices provide real-time valuation reference	No observable market price, requires estimation based on various methods
Ownership Structure	Dispersed ownership, professional management, governance standards	Concentrated ownership, often owner-operated with potential key person risks
Liquidity	Highly liquid shares traded daily on exchanges	Illiquid shares with no ready market, requiring discounts of 20-30%

MARKET VALUE NOT AVAILABLE

1. Computing Beta
2. Debt ratios for going from unlevered to levered betas
3. Computing cost of capital
4. Compute the value of options and warrants granted to employees
5. Compute volatility to estimate the option value

CASH FLOW ESTIMATION ISSUES

- ◉ Different Accounting Standards
- ◉ Personal and business expenses
- ◉ Salaries from dividends
- ◉ Pass-through entities: no entity-level tax for proprietorships/partnerships
 - changes after-tax cash flow and tax-shield estimates

ADJUSTMENTS IN FINANCIAL STATEMENTS

- ◉ Profit figures might be overstated.
- ◉ Assets: Personal and Business Assets
 - Cars, Office (if home used)
- ◉ Private company financials often need adjustments for owner compensation, personal expenses, and non-recurring items
 - Salary for ghost employees, personal travel expenses, attending seminars, high salary for the CEO

CONTROL PREMIUMS

- ◉ Control premium: 20-30% premium for the ability to direct operations and strategy
- ◉ Minority discount: 15-25% reduction for lack of control over decisions
- ◉ *Ranges reflect empirical control-premium and minority-discount studies (e.g., FactSet Mergerstat Control Premium Study).*

DISCOUNT RATE

Why Discount Rates Differ for Private Companies

- ◉ Higher risk profile: Increased uncertainty in forecasts and operations.
- ◉ Concentrated ownership: Non-diversified owners bear additional unsystematic risk

Calculating Cost of Equity

- ◉ Build-up method: Risk-free rate + equity risk premium + additional premiums
- ◉ Size premium: Additional 3-5% for small or micro-cap companies
- ◉ Illiquidity factor: Higher discount rates to reflect lack of marketability

Ranges follow build-up method literature (Duff & Phelps/Kroll Cost of Capital Navigator; Ibbotson SBBI size-premium studies).

PRIVATE COMPANY VALUATIONS: THREE POSSIBLE SCENARIOS

- ◉ Private to private transactions
- ◉ Private to public transactions
- ◉ Private to IPO

PRIVATE TO PRIVATE TRANSACTION

1. Neither the buyer nor the seller is diversified.
2. The investment is illiquid.
3. Key person value.

KEY PERSON VALUE

- The value of the business is inversely proportional to the stature of the key person in the company.
- The buyer buys the company *minus* the key person.
- If the company can survive without the key person, the value of the business will be high.

AN EXAMPLE: VALUING KAVITA TEXTILES

- Private Company selling textile garments in North India.
 - 60% revenue comes from selling school uniforms.
 - 40% revenue comes from selling ready-made garments in shopping outlets.
 - About 100 workers - mostly part-time work and stitch the garments using **single-needle manually-operated sewing machines**.
- The company wants to grow and needs **digital control multi-needle quilting machines**.
 - Requires Rs.75 crores. Rs.50 Crores for buying the machines, building a new factory, etc. and Rs.25 crores for working capital financing.
 - A PSU Bank has offered to lend Rs.25 crores at 11%, provided the remaining amount comes in equity.

FINANCIALS OF KAVITA TEXTILES PRIVATE LIMITED

Profit and Loss Account of Kavita Textiles Private Limited (Figures in ` Crores)*      *Balance Sheet of Kavita Textiles Private Limited (Figures in ` Crores)

<i>For the year ending 31 March of</i>	<i>2011</i>	<i>2012</i>	<i>As on 31 March of</i>	<i>2011</i>	<i>2012</i>
Net Sales	20	26	Loan from Bank	5	5
Other Income	1	1	Share Capital	3	3
Total Revenue	21	27	Reserves and Surpluses	10	12.6
Total Expenses			Total Liabilities	18	20.6
Raw Material	8	10.5	Cash in Bank	5	7
Wages and Salaries	1	1.2	Accounts Receivables	1	2
Selling and Distribution Expenses	1	1.5	Inventories	8	9
Administrative Expenses	1.2	1.3	Loans and Advances	1	1
Depreciation	1.1	1.1	Total Current Assets	15	19
Total	12.3	15.6	Accounts Payable	2	4
Interest	2	2.4	Provisions	1	1
Provision for Tax	1.7	2.3	Total Current Liability	3	5
Profit after Tax	5.0	6.8	Net Current Assets	12	14
			Gross Fixed Assets	8	9.7
			Less: Accumulated Depreciation	2	3.1
			Net Fixed Assets	6	6.6
			Total Assets	18	20.6

Cash tax rate: 25%

DIFFERENCES IN THE FINANCIAL STATEMENTS OF PUBLIC AND PRIVATE COMPANIES

- Kavita and her husband currently draw salary of Rs.100,000 each p.m. For an equivalent public company, the salary would be Rs.500,000.
 - Revised wages and salaries expenses will increase to $\text{Rs.1.2 cr} + 400,000 \times 12 \times 2 = \text{Rs.2.16 crores}$.
- Selling and Distn expenses and admin. Expenses are probably overstated as they might include some personal expenses. I restate at Rs.2 crores.

REVISED FINANCIALS

Profit and Loss Account of Kavita Textiles Private Limited (Figures in ` Crores)

<i>For the year ending 31 March of</i>	2011	2012	2012-Revised
Net Sales	20	26	26
Other Income	1	1	1
Total Revenue	21	27	27
Total Expenses			
Raw Material	8	10.5	10.5
Wages and Salaries	1	1.2	2.16
Selling and Distribution Expenses	1	1.5	1
Administrative Expenses	1.2	1.3	1
Depreciation	1.1	1.1	1.1
Total	12.3	15.6	15.76
Interest	2	2.4	2.4
Provision for Tax	1.7	2.3	2.2
Profit after Tax	5.0	6.8	6.6

ESTIMATING DISCOUNT RATES

Company Name	Cash	Debt	MktCap (` Crores)	P/E	P/B	Enterprise value (EV) (` Crores)	EV / EBITDA	Beta
Spice Islands Apparels Ltd.	47.8	15	51.13	8.52	0.35	66.13	0.11	0.56
Virat Industries Ltd.	10.8	14.8	80.25	5.39	0.87	95.05	3.94	0.54
Cityman Ltd.	0.1	119.5	125.9	NA	NA	245.4	NA	0.85
G I V O Ltd.	20.5	36.9	173.07	37.62	0.6	209.97	13.44	1.19
Samtex Fashions Ltd.	11.4	439.1	173.25	91.18	0.42	612.35	9.79	0.73

Company Name	Cash	Debt	MktCap (` Crores)	Beta	Net DE	Beta-U
Spice Islands Apparels Ltd.	47.8	15	51.13	0.56	-0.64	1.08
Virat Industries Ltd.	10.8	14.8	80.25	0.54	0.05	0.52
Cityman Ltd.	0.1	119.5	125.9	0.85	0.95	0.50
G I V O Ltd.	20.5	36.9	173.07	1.19	0.09	1.11
Samtex Fashions Ltd.	11.4	439.1	173.25	0.73	2.47	0.26
Median				0.73	0.09	0.52

ESTIMATING K_E OF KTL

CAPM build-up: $K_e = R_f + \text{Beta} \times \text{Equity Risk Premium}$

Market Value of Equity = $17 \times \text{Rs.}6.6 \text{ crores} = \text{Rs.}112.71 \text{ crores}$.

Here, 17 is the median Earnings multiple and Rs.6.6 crores is the adjusted net income of the company.

$$\begin{aligned}\beta_L &= \beta_U \times \left(1 + (1 - t) \times \frac{D}{E} \right) \\ &= 0.52 \times \left(1 + 0.75 \times \frac{-2}{112.71} \right) \\ &= 0.51\end{aligned}$$

$$K_e: 8.2\% + 0.51 \times 7.13\% = \mathbf{11.84\%}.$$

BUT KAVITA IS NOT DIVERSIFIED...

- ◉ Finance theory simply assumes people like Kavita to be irrational.
- ◉ Typical shortcuts used:
 - Double the beta: K_e will equal 15.47% (vis-à-vis 11.84%)
 - Double the K_e : K_e will equal 23.67%
 - Use Market Model

$$r_{i,t} = \alpha_i + \beta_i \times r_{m,t} + errors$$

$$\sigma_i^2 = \beta_i^2 \times \sigma_m^2 + \sigma_{errors}^2$$

ESTIMATING NON-SYSTEMATIC RISK

Company Name	Total Risk	Systematic Risk	Non-systematic Risk
Spice Islands Apparels Ltd.	37.71%	3.07%	34.63%
Virat Industries Ltd.	31.67%	2.86%	28.82%
Cityman Ltd.	31.20%	7.08%	24.12%
G I V O Ltd.	37.84%	13.88%	23.96%
Samtex Fashions Ltd.	51.68%	5.22%	46.45%

- ◉ Non-systematic risk around 5 times systematic risk.
- ◉ Adjusted K_e will equal 71%.
- ◉ I use 23.67% here.

COST OF CAPITAL AND EQUITY VALUE

- ◉ Cost of Debt: 11%
- ◉ Cost of Equity: 23.67%
- ◉ Tax Rate: 25%

$$WACC = \frac{-2}{-2 + 112.7} \times 0.11 \times (1 - 0.25) + \frac{112.7}{-2 + 112.7} \times 0.2367 = 23.95\%$$

Free Cash Flow	7.68	7.26	9.53	12.25	14.90	17.23
Terminal Value						135.91
PV of FCF	77.15					
Less: Debt	5					
Add Cash	7					
Equity Value	79.15					

The value (ignoring Non-systematic risk) would have been Rs.459.38 crores.

ONE MORE PROBLEM: KAVITA'S SHARES ARE NOT LISTED.

- They are illiquid. Investors take illiquidity risk.
 - Not a diversifiable risk. So (the risk) deserves a premium.
- Using 25% discount, the value of Kavita's equity comes to $0.75 \times \text{Rs.}79.15 \text{ crores} = \text{Rs.}59.36 \text{ crores}$.
 - Not bad for (reported) net worth of Rs.15.6 crores.

HOW IS THE ILLIQUIDITY DISCOUNT (DLOM) ACTUALLY ESTIMATED?

- **DLOM: Discount for Lack of Marketability**
- **Restricted stock studies:** discounts on privately placed shares of otherwise-public companies, typically 20-35%
- **Pre-IPO studies:** price appreciation between a private placement and the subsequent IPO, often 30-50% historically (narrower in recent samples)
- **Option-pricing (protective put) models:** cost of a put option that hedges the holding period until a sale or IPO is possible
- **In KTL's case:** we used a flat 25% - within the empirical range, but a judgment call rather than a model output. Ask: which method would you defend to a client, and why?
- **Sensitivity:** on the same Rs.79.15cr pre-discount base, a 15% DLOM gives Rs.67.28 crores while a 30% DLOM gives Rs.55.41 crores - a swing of nearly Rs.12 crores from this one assumption alone

SENSITIVITY CHECK: HOW MUCH DOES THE KE CHOICE MATTER?

Scenario	Ke used	Resulting equity value
Diversified investor (plain CAPM)	11.84%	Rs.459.38 crores
Undiversified, "double the beta" shortcut	15.47%	Compute
Undiversified, "double the Ke" shortcut (used in case)	23.67%	Rs.79.15 crores (pre-DLOM)
Undiversified, total (non-systematic) risk approach	71%	Compute

A single subjective call on how to price Kavita's lack of diversification swings the pre-illiquidity equity value by roughly 6x (Rs.459.38cr vs Rs.79.15cr) - before the illiquidity and growth adjustments are even applied.

VALUING KTPL UNDER GROWTH ASSUMPTION

- ◉ Expanding beyond north India. Growth will slow down in the beginning.
- ◉ KTPL has plans of using automated machines for doing a large part of the stitching.
 - The wages and salaries expenses will not increase in line with the sales.
- ◉ Need to hire more managers to take care of operations. Private investors look for
 - COO, Director (Marketing) and Auditor (CA)
- ◉ Selling and distribution expenses will be high in the beginning.

PROJECTED FINANCIALS

<i>Profit and Loss Account of Kavita Textiles Private Limited (Figures in ` Crores)</i>						
<i>For the year ending 31 March of</i>	2012-Revised	2013E	2014E	2015E	2016E	2017E
Net Sales	26	50.00	80.00	120.00	150.00	180.00
Other Income	1	6.15	2.69	4.26	6.28	9.38
Total Revenue	27	56.15	82.69	124.26	156.28	189.38
Total Expenses						
Raw Material	10.5	20.00	32.00	48.00	60.00	72.00
Wages and Salaries	2.16	3.82	5.53	7.77	9.61	11.54
Selling and Distribution Expenses	1	3.00	4.80	4.56	5.70	6.84
Administrative Expenses	1	1.90	3.04	4.56	5.70	6.84
Depreciation	1.1	5.27	8.56	7.96	7.47	7.06
Total	15.76	33.98	53.92	72.86	88.48	104.27
Interest	2.4	5.15	5.15	5.15	5.15	5.15
Provision for Tax	2.2	4.3	5.9	11.6	15.7	20.0
Profit after Tax	6.6	12.8	17.7	34.7	47.0	60.0

PROJECTED FREE CASH FLOW

Free Cash Projections for Kavita Textiles Private Limited (Figures in ` Crores)						
	2012-Revised	2013E	2014E	2015E	2016E	2017E
Net Operating Income	10.24	16.02	26.08	47.14	61.52	75.73
Less: Adjusted Tax	2.56	4.00	6.52	11.79	15.38	18.93
NOPAT	7.68	12.01	19.56	35.36	46.14	56.79
Add: Depreciation	1.10	5.27	8.56	7.96	7.47	7.06
Gross Cash Flow	8.78	17.28	28.11	43.32	53.61	63.85
Capex	1.10	50.00	5.00	5.00	5.00	5.00
Investment in Non-cash working capital	0.00	14.15	0.37	10.76	8.07	8.07
Gross Investment	1.10	64.15	5.37	15.76	13.07	13.07
Free Cash Flow	7.68	-46.87	22.74	27.56	40.54	50.78

NEW COST OF CAPITAL

- ◉ Debt ratio has changed now.
 - New Debt: Rs.30 crores. Net debt: Rs.23 crores. New Equity: Rs.112.71 (59.36?) + Rs.50 crores.
 - Net Beta = 0.575
 - *Re-levering formula: $Beta-L = Beta-U \times [1 + (1 - Tax Rate) \times D/E]$*
 - New Ke = 12.3%
- ◉ New investors may be diversified.
 - I use 20% as the Ke here.
- ◉ Cost of equity is a negative function of size of the company (Fama and French, 1993)

$$WACC = \frac{23}{23 + 162.71} \times 0.11 \times (1 - 0.25) + \frac{162.71}{23 + 162.71} \times 0.2 = 17.52\%.$$

FINAL EQUITY VALUE

- ◉ Implied Equity Value: Rs.345.67 Crores
- ◉ Illiquidity Adjustment: $\text{Rs.345.67 crores} \times (1 - 0.25) = \text{Rs.259.25 crores}$.
- ◉ This is the post-money value.
- ◉ Pre-money value is Rs.209.25 crores.
- ◉ 30 Lakh shares outstanding.
- ◉ Implied value per share = Rs.864 per share.
- ◉ New investors will be issued 0.06 crore shares.

ADJUSTED BALANCE SHEET

<i>Balance Sheet of Kavita Textiles Private Limited (Figures in ` Crores)</i>						
<i>As on 31 March of</i>	2012-Revised	2013E	2014E	2015E	2016E	2017E
Loan from Bank	30	30	30	30	30	30
Share Capital	3.6	3.6	3.6	3.6	3.6	3.6
Reserves and Surpluses	62	74.8	92.5	127.2	174.1	234.1
Total Liabilities	95.60	108.36	126.08	160.77	207.75	267.71
Cash in Bank	82	35.88	56.78	83.67	125.05	179.01
Accounts Receivables	2	3.85	6.16	9.24	11.55	13.86
Inventories	9	25.00	27.68	41.52	51.90	62.28
Loans and Advances	1	1.95	3.12	4.68	5.85	7.02
Total Current Assets	94	66.68	93.74	139.11	194.35	262.17
Accounts Payable	4	7.70	12.32	18.48	23.10	27.72
Provisions	1	1.95	3.12	4.68	5.85	7.02
Total Current Liability	5	9.65	15.44	23.16	28.95	34.74
Net Current Assets	89	57.03	78.30	115.95	165.40	227.43
Gross Fixed Assets	9.7	59.70	64.70	69.70	74.70	79.70
Less: Accumulated Depreciation	3.1	8.37	16.92	24.89	32.35	39.41
Net Fixed Assets	6.6	51.33	47.78	44.81	42.35	40.29
Total Assets	95.60	108.36	126.08	160.77	207.75	267.71

ASSESS THE IMPACT OF THE KEY PERSON

- Unlikely to affect value here as both Kavita and her husband will continue in this business.
- In a normal case, if the key person leaves:
 - Enter into a no-compete agreement
 - Key-man insurance to fund a buyout or transition
 - Build management depth so operations do not depend on one person
 - Structure the deal with an earn-out or escrow tied to the key person's retention

PRIVATE COMPANY FOR IPO

- ⦿ The investors are clearly diversified.
- ⦿ The reporting and information disclosure requirements shift to reflect a publicly traded firm.

USE OF THE PROCEEDS

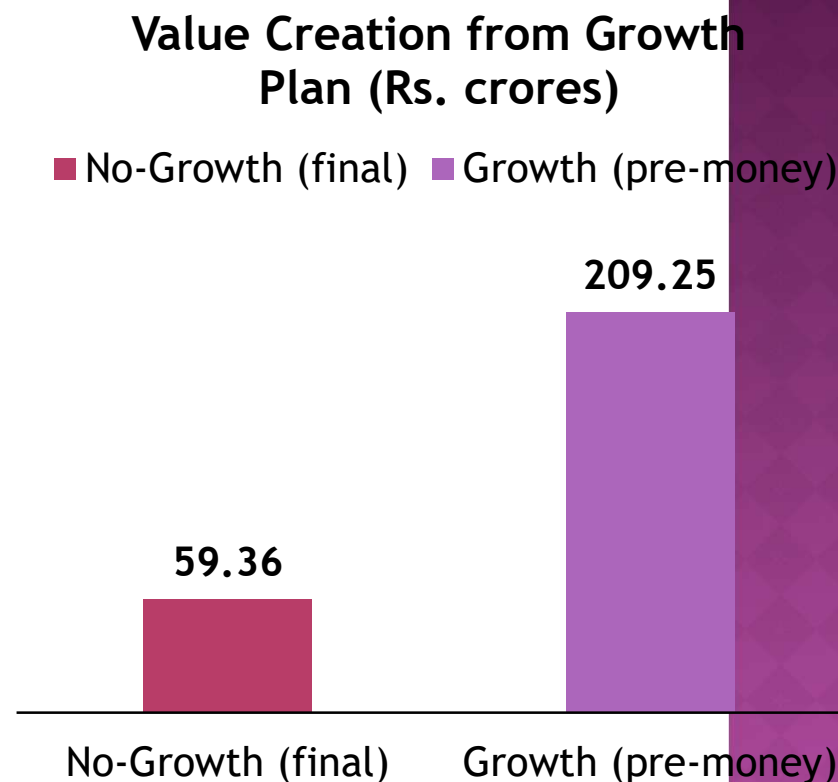
- ◉ The proceeds from an IPO can be
 - Taken out of the firm by the existing owners
 - Used to pay down debt and other obligations
 - Held as cash by the company to cover future reinvestment needs
- ◉ How you deal with the issuance will depend upon how the proceeds are used.
 - If taken out of the firm -> ignore in valuation
 - If used to pay down debt -> Change the debt ratio, which may change the cost of capital and the value of the firm
 - If held as cash to cover future reinvestment needs -> add the cash proceeds from the IPO to the DCF valuation of the company.

EXTENDING THE CASE: IF KTL CHOSE AN IPO INSTEAD

- Growth-plan private placement value: Rs.259.25 crores post-money (Rs.864/share), after a 25% illiquidity discount on Rs.345.67 crores
- **No more DLOM:** listed shares are marketable, so value should move back up toward the undiscounted Rs.345.67 crores
- **Diversified investor base:** IPO investors are diversified, so K_e should move toward the plain CAPM figure of 11.84% rather than the 20-23.67% used for a single undiversified owner - pushing value higher still
- **Use of the Rs.50 crore raised matters:** if it funds the planned expansion (as modeled), it stays inside the DCF; if any portion is instead taken out by Kavita's family, that portion must be excluded from firm value

KAVITA TEXTILES: VALUE BRIDGE ACROSS SCENARIOS

Step	Equity value
Market-multiple value (17x earnings)	Rs.112.71 crores
Fully diversified investor (Ke = 11.84%)	Rs.459.38 crores
No-growth, undiversified + illiquid (final)	Rs.59.36 crores
Growth plan, pre-illiquidity (Ke = 20%)	Rs.345.67 crores
Growth plan, post-money (final)	Rs.259.25 crores
Growth plan, pre-money	Rs.209.25 crores



Value creation from the growth plan: ~3.5x (Rs.209.25cr pre-money vs Rs.59.36cr no-growth)

PRIVATE COMPANY VALUATION: CLOSING THOUGHTS

- ⊙ The value of a private business will depend on the potential buyer (diversification, time horizon, need for cash, etc.)
- ⊙ If you are a seller of a private business, you will maximize value, if you sell to
 - One who is well-diversified (or whose investors are)
 - And who does not think too highly of you (as a person)